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Safe System Circle – A Novel Approach in HSE Culture Transformation for Deepwater Exploration Drilling Campaign

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Abstract

Fostering a strong Health, Safety, and Environment (HSE) culture during a short-term offshore deepwater drilling campaign presents considerable challenges—particularly due to the condensed project timeline and the intensity of high-risk operations involved. Despite these constraints, the Drilling HSE team was entrusted with a critical mission: to support the company's first deepwater drilling campaign while achieving Zero Lost Time Incidents (LTI), a key HSE performance indicator.

To meet this strategic objective, the team launched an innovative initiative titled "Safe System Circle (SSC)", designed to accelerate cultural transformation. The SSC model integrates elements of the company's Safe System of Work (SSOW), quality circle methodology, and a structured HSE coaching framework to align frontline safety behaviors with the organization's overarching safety values.

Studies indicate that 80-90% of losses in industrial operations stem from human error, both at the individual and organizational levels. Despite this, only 3.4% of organizations have established mature systems for proactively identifying, evaluating, and managing risks. A positive safety culture is increasingly recognized as a prerequisite for strong safety performance and often serves as a reliable predictor of such outcomes. However, cultivating a positive safety culture typically requires significant time and sustained effort, which may not be feasible for short-duration projects such as exploration drilling campaigns.

In response, the Drilling HSE team in Indonesia adapted a fundamental coaching strategy into the SSC framework, emphasizing peer engagement and hazard awareness through structured safety discussions. The initiative was spearheaded by the onshore HSE support team, who developed a customized coaching curriculum specifically tailored for offshore deployment on a drillship. The program was rolled out in phases, beginning with the communication of essential safety principles aligned with SSOW to a core team onboard. This was followed by targeted safety leadership training for key personnel to build ownership and reinforce accountability. Finally, a tailored Behavior-Based Safety Observation (BBSO) component was implemented to drive continuous improvement in frontline practices.

The case study presented in this paper highlights how a focused and inclusive approach—centered on workforce participation and behavioral change, can meaningfully enhance safety culture within a limited timeframe. The SSC model was successfully applied across three of the company's offshore exploration

drilling campaigns, contributing significantly to the achievement of the Zero LTI target and demonstrating its value as a replicable tool for future projects.

Introduction

The Exploration Drilling Project is located in the offshore deepwater area of the, Aceh Province. The distance from the well locations to the Krueng Geukueh Shore Base in Lhokseumawe is approximately 52 nautical miles (7–8 hours by Platform Supply Vessel) for Well #1, and 36 nautical miles (5–6 hours) for Well #2. Drilling operations are carried out using a Drillship, with an estimated water depth of around 1,188 meters at the site.

Helicopter support is provided from the Sultan Iskandar Muda Airport in Banda Aceh, which serves as the Heli Base. The air distance from the Heli Base to the drilling locations is approximately 109 nautical miles (60 minutes flight time) for Well #1 and 90 nautical miles (55 minutes) for Well #2.

As part of the company's commitment to operational excellence, all company leaders, including those in site management, are expected to:

- Demonstrate and reinforce positive HSE behaviors both on and off duty.
- Actively engage with employees, contractors, and third parties to promote excellence in the Operations Management System (OMS) and related standards.
- Support the development and implementation of OMS, ensuring documented systems are in place to demonstrate compliance.
- Integrate HSSE, functional, and common requirements into business strategies, work programs, and daily operations.
- Involve employees meaningfully in the planning, execution, and review of HSE and functional requirements.
- Ensure adequate resources are allocated to meet each requirement.
- Assign clear HSE and functional responsibilities where applicable and assess employees against their annual objectives.
- Foster the sharing of HSE-related lessons learned across and beyond the Business Unit.

In line with these leadership expectations, the Drilling Management Team (DMT) of the Indonesia Business Unit (IBU), in collaboration with the HSE team.

Sustaining strong HSE performance is of utmost importance, particularly as IBU has maintained a 12-year record of zero Lost Time Incidents (LTI), a record the team is fully committed to upholding. This campaign also marks a significant milestone as the first offshore deepwater drilling operation in the company's history, further emphasizing the need for heightened focus on safety and operational integrity.

The company is committed to working with our Contractor Companies to form a partnership that strives for excellence in HSSE performance throughout the execution of the Exploration Drilling project in Q1 and Q2 year 2024. The responsibility for implementing the HSE Golden Rules is not only limited to the Company and the Contractor but must involve subcontractors' line management. Every group leader must be responsible to ensure the HSSE Golden Rules are implemented as per company requirement.

The Contractor Companies will commit to put their resources and team to implement the "HSE Golden Rules" standard to achieve a project free of fatalities, injuries, occupational illness, harm to the environment, loss of property and any other significant incidents.



Figure 1—HSE Golden Rules

Methodology

To drive effective improvement in HSE culture, a novel methodology was developed that integrates the Rig HSE coaching model, the company's Safe System of Work (SSOW) standard, and the principles of the Quality Circle Model (QCM). Traditionally, QCM involves groups of employees meeting regularly to solve problems and improve operations. This approach was adapted to focus on HSE and rebranded as the Safe System Circle (SSC). The SSC initiative was led by the Onshore HSE Team, which developed an HSE coaching syllabus tailored for offshore drillship implementation. As the program progressed, safety performance showed marked improvement within a relatively short timeframe. The approach began with establishing and communicating core safety principles, in alignment with SSOW, to a small team onboard the rig. This was followed by delivering safety leadership training to key personnel to ensure understanding and commitment. Lastly, a behavior-based safety observation (BBSO) program was introduced and customized to suit the offshore work environment.

The overall methodology will be as follows:

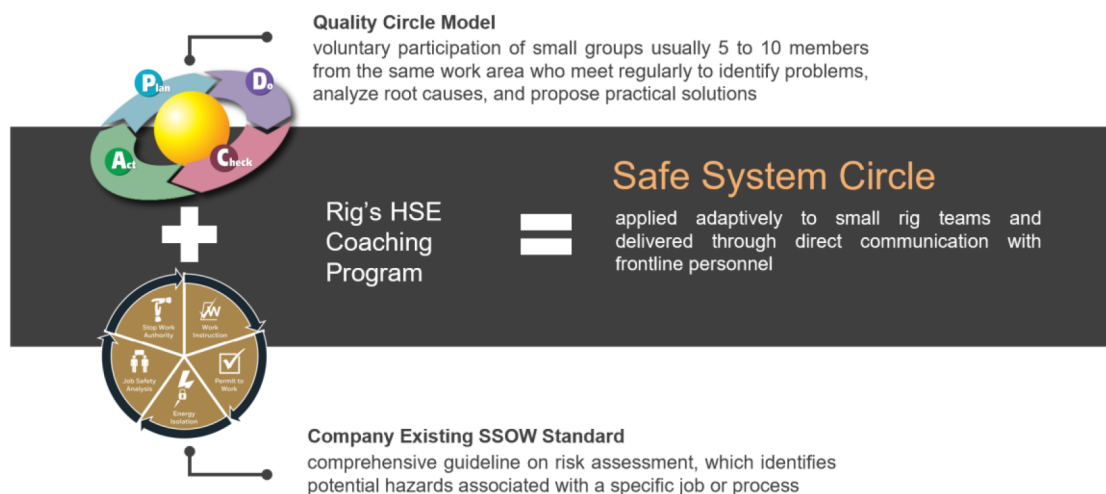


Figure 2—Concept Methodology

Rig HSSE Coach Program

The Rig HSE Coach Program is a structured initiative designed to enhance Health, Safety, and Environment (HSE) performance on drilling rigs through on-site coaching, behavioral reinforcement, and proactive hazard management. Unlike traditional compliance-based inspections, the HSE Coach Program focuses on

building safety leadership, developing frontline ownership, and promoting a positive safety culture. It is widely applied in high-risk environments such as offshore and onshore drilling operations, where human behavior plays a significant role in preventing incidents (Paté-Cornell, 2009).

A key component of the program is behavior-based observation and coaching. HSE Coaches, who are often seasoned safety professionals or trained operational leaders, spend time on the rig floor observing work practices, identifying at-risk behaviors, and engaging workers in constructive, non-punitive conversations. These interactions help reinforce safe behaviors, clarify expectations, and correct unsafe habits in real-time. According to Cooper (2015), such engagement has a direct impact on safety performance by bridging the gap between safety procedures and actual worker behavior.

The success of a Rig HSE Coach Program depends on several factors: the credibility of the coach, the consistency of message delivery, and the ability to build trust with the rig crew. Rather than acting as enforcers, effective HSE Coaches position themselves as mentors and facilitators who empower workers to take ownership of safety. This aligns with modern safety thinking that emphasizes "safety as a capacity" rather than merely the absence of incidents (Dekker, 2014).

Additionally, the program often includes data collection and trend analysis, where coaches document observed behaviors, categorize findings, and provide feedback to both rig teams and management. This feedback loop enables the organization to identify systemic issues, develop targeted interventions, and measure progress over time. As noted by Krause and Hidley (2013), such structured observation programs not only reduce incident rates but also drive continuous improvement by making safety visible and personal.

In summary, the Rig HSE Coach Program is more than a safety initiative—it is a leadership and cultural change tool that transforms the way safety is perceived and practiced on the rig. It reinforces accountability, supports frontline leadership, and builds a resilient workforce capable of anticipating and managing operational risks.

Safe System of Work

A Safe System of Work (SSOW) is a formal procedure that outlines how tasks should be carried out to minimize the risk of injury or harm to workers and others in the workplace. It is particularly important for tasks involving significant hazards where risks cannot be fully eliminated through design or engineering controls alone. SSOWs serve as the operational foundation of a company's risk management strategy, ensuring that work is planned, authorized, and executed in a manner consistent with both legal requirements and good safety practice (Hughes & Ferrett, 2020).

The development of an SSOW typically begins with a comprehensive risk assessment, which identifies potential hazards associated with a specific job or process. Based on this assessment, the SSOW outlines a series of control measures—technical, procedural, and organizational—that workers must follow. This may include isolation procedures, permit-to-work systems, use of personal protective equipment (PPE), and supervision requirements. According to Ridley and Channing (2013), the effectiveness of an SSOW depends on the clarity of its instructions, the competence of personnel involved, and the adequacy of training provided to those carrying out the tasks.

One common framework within SSOWs is the Permit-to-Work (PTW) system, which is used to control high-risk activities such as hot work, confined space entry, or work at height. The PTW process ensures that all necessary precautions have been taken before work begins and that communication between departments is maintained throughout. When integrated properly, SSOWs and PTWs can reduce the likelihood of human error and improve overall safety performance (Health and Safety Executive, 2019).

However, SSOW is not a one-time document. It must be dynamic and regularly reviewed to reflect changes in working conditions, equipment, or personnel. Worker feedback is crucial in this regard, as frontline employees often have firsthand insight into practical risks and implementation issues. As Gunningham (2008) argues, safety systems are only effective when embedded within an organizational

culture that values transparency, learning, and shared responsibility. In this sense, SSOWs are both a procedural and cultural tool in the pursuit of workplace safety.

Quality Circle Model

The QCM through organized group activities. At its core, the model revolves around voluntary participation of small groups usually 5 to 10 members from the same work area who meet regularly to identify problems, analyze root causes, and propose practical solutions. The model typically follows a systematic process, often visualized in a cycle that includes problem identification, data collection, cause analysis, solution generation, implementation, and follow-up (Ishikawa, 1985).

A key feature of the QCM is its integration with problem-solving tools and techniques such as cause-and-effect diagrams (Ishikawa diagrams), Pareto analysis, control charts, and brainstorming. These tools guide teams in performing structured and data-based analysis rather than relying on assumptions. The model encourages using the Plan-Do-Check-Act (PDCA) cycle to implement and monitor solutions, ensuring that changes are tested and refined before full-scale adoption (Deming, 1986).

Another essential element of the model is facilitation and support by middle management. While quality circles are grassroots in nature, they function effectively when there is institutional backing, including training in quality tools, recognition of team achievements, and alignment with organizational goals. According to Sharma and Talib (2015), organizations that embed quality circles into their operational strategy often experience improved teamwork, enhanced communication, and measurable performance gains.

The QCM can be adapted to different organizational settings and cultures. In Japan, where the concept originated, the model emphasizes consensus and group harmony, whereas Western adaptations may place more emphasis on individual accountability and innovation. Despite these variations, the foundational structure of the model remains consistent: a cycle of continuous employee-driven improvement supported by management systems and quality techniques (Ross & Ross, 2006).

Behavioral Based Safety Observation

Behavior-Based Safety Observation (BBSO) is a proactive safety management approach that focuses on identifying and reinforcing safe behaviors while correcting at-risk behaviors through structured observations and feedback. Rooted in behavioral psychology, the premise of BBSO is that most workplace incidents are caused by human actions, and by observing and influencing these behaviors, organizations can significantly reduce incident rates (Geller, 2005). Rather than relying solely on rules and procedures, BBSO emphasizes coaching and peer involvement to create lasting behavioral changes.

The process typically involves trained observers—either supervisors or peer employees—who systematically monitor work activities using customized checklists that focus on critical behaviors relevant to the task or environment. Observations are non-punitive and are followed by immediate feedback, often in the form of positive reinforcement or constructive conversation. According to Krause et al. (1990), this immediate feedback loop increases workers' awareness of their actions and helps embed safety habits into routine practice.

An essential element of successful BBSO programs is data collection and analysis. Observation results are aggregated to identify behavior trends, common at-risk actions, and systemic issues. These insights are then used to develop targeted interventions such as training, procedure revisions, or engineering controls. The use of data-driven decision-making ensures that safety improvements are evidence-based rather than anecdotal (Cooper, 2009). Additionally, involving workers in the development of observation checklists and the review of observation data enhances ownership and buy-in.

Despite its proven benefits, BBSO programs must be carefully implemented to avoid pitfalls such as mistrust, token compliance, or "checklist fatigue." The success of BBSO depends heavily on organizational culture, particularly trust, openness, and visible management commitment. Dekker (2011) cautions that

behavior-based programs should not become a substitute for addressing systemic risks or engineering failures; rather, they should complement a comprehensive safety system that includes technical, procedural, and cultural components.

In conclusion, Behavior-Based Safety Observation is a valuable strategy for building a strong safety culture by aligning individual behaviors with organizational safety goals. When executed thoughtfully, it enhances communication, empowers frontline workers, and creates a positive feedback environment that sustains continuous improvement in safety performance.

Delimitations

This paper focuses on the implementation of an approach that combines various safety culture tools for use on a deepwater drilling rig during an exploration campaign. The SSC was implemented on one rig and operated under the company's SSOW standard for approximately three months. This experience is intended to inform the application of similar programs in future deepwater drilling campaigns.

Novel Approach Implementation

This paper presents a novel method for Health, Safety, and Environment (HSE) implementation, designed to strengthen the safety culture throughout a deep-water exploration drilling campaign in Indonesia. The SSC methodology was adaptively employed with small rig teams and communicated directly to frontline personnel.

Based on the conceptual approach previously discussed, the team has identified the following initiatives as part of a novel strategy to enhance HSE culture within the project:

- Securing support from the Indonesia Business Unit (IBU) and Drilling Department Management teams to deploy HSE coaches, as well as an additional Rig HSE Supervisor, to ensure comprehensive 24-hour coverage for rig operations.
- Strengthening accountability and oversight in HSE practices by developing a program that positions the Drilling Site Manager (DSM) as an active leader responsible for achieving HSE objectives throughout the drilling campaign.
- Engaging an external HSE coach provider to facilitate coaching and training, identify key focus areas, establish KPIs, and design an evaluation framework.
- Conducting regular weekly and monthly reviews of SSC implementation, with outcomes used to inform ongoing improvement efforts.



Figure 3—Novel Approach Flow

Management Commitment

This deepwater drilling campaign marks the company's first such operation. The project commenced amid heightened concern following an incident during the rig's relocation from the previous concession area to the company's designated site, where a Lost Time Incident (LTI) occurred resulting in a broken left arm due to a fall attributed to housekeeping issues. Management has expressed concerns regarding the prevailing HSE culture on the rig, particularly due to a potential Lost Time Incident (LTI) that occurred during exploration drilling operations for another operator. The company has not recorded any LTIs since 2013 and aims to maintain this level of performance. The HSE and Drilling departments have been assigned to identify potential actions to further improve the HSE culture and prevent incidents when the rig is under the company's control.

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The HSE and Drilling departments decided to explore innovative approaches to further strengthen the HSE culture onboard the Drillship throughout the exploration drilling phase, which spans approximately six months by implementing SSC with the measurable target as follows:

- Safety Conversations / Leadership Engagement
- Hazard Identification, Behavior-Based Safety (BBS) Observations and Reporting
- Emergency Drills and Preparedness
- Permit to Work & Risk Assessment Audit
- Injury/Incident Rates and Near Miss Reporting
- Environmental Program Matrix Implementation
- Feedback Mechanism Utilization

The proposal to implement SSC as outlined above received full endorsement from management, including authorization to engage an external HSE Coach consultant, approval to increase the number of Rig HSE Coordinators to ensure round-the-clock coverage, and agreement to involve additional HSE support from the company's production HSE team. Management further approved active participation by DSM in rig HSE implementation; this includes requiring DSM to submit weekly reports on SSOW implementation using a Site Leadership Card, with direct reporting to the IBU Management Team.

Table 1—Site Leadership Card

Process	Actions	Follow Up Requirements
Work planning and scheduling	LAH Program Identification and PTW Required	Every 3 days submitted to Drilling Supt. and Drilling Manager Summary of High-Risk Activities requiring PTW, SIMOPS and specific competencies
Understanding the scope of the task	Work scope communication to affected parties	Every 3 days submitted to Drilling Supt and Drilling Manager sample evidence of PTW audit conducted by DSM and OIM especially to High-Risk activities including findings and follow-up
HAZID and risk management	Review PTW, JSA and Risk Management process	Every 3 days submitted to Drilling Supt & Drilling Manager sample evidence of PTW audit conducted by DSM and OIM especially to High Risk Activities including findings and follow up
Control Implementation	Implementing mitigation and action as per PTW, JSA and Risk Management process	Every 3 days submitted to Drilling Supt & Drilling Manager sample evidence of follow up from PTW implementation (isolations, hot work, confined space, working at height)
Monitoring Control	Competent person periodically visits and inspects the work site to ensure conditions detailed in the risk assessment	Every 3 days submitted to evidence of Hazard Hunts follow up and closure plan. Prepare time to have a meeting/discussion with Drilling Supt & Drilling Manager on the control implementation
Auditing	Conduct random sampling audit of PTW and follow up site the implementation	Every 3 days submitted to Drilling Supt & Drilling Manager sample evidence of PTW audit conducted by DSM and OIM especially to High Risk Activities including findings and follow up
Continuous Improvement	Review HOC, inspection result and HSSE related training implementation	Every 7 days submitted to Drilling Supt & Drilling Manager sample evidence of Review HOC, inspection result and HSE related training implementation including action tracker

Planning and Execution of SSC

Following management approval, the onshore team engaged a consultant who had previously delivered comparable services for another company's deepwater drilling campaign with the same drilling contractor. This strategy enables the onshore team to identify common HSE concerns encountered in prior campaigns on the same rig and to tailor their approach in alignment with the company's defined measurable objectives.

To support the company's SSOW standard implementation, the onshore team decided to integrate the consultant's HSE coach program with SSOW-specific requirements during a short-term drilling campaign, using the quality circle approach commonly seen in manufacturing. This method involved dividing rig crews into smaller groups to facilitate effective coaching sessions and discussions about potential hazards relevant to each group, as well as strategies for mitigating these hazards both within the group and across related groups.

Onshore team planning related to coach scheduling, program development, definition of delivery methods, identification of circles, and selection of evaluation criteria was conducted on an accelerated timeline to align with rig arrival and as much as possible, the drilling schedule. Following discussions with IBU Management, the consultant and onshore team agreed on the following:

- The rig team is divided into smaller task groups (maximum 5 persons) assigned to specific Job Safety Analyses (JSA), with prior coordination with Rig Management and inclusion in rig safety meetings.
- Based on the JSA, coaches determine themes for specific tasks, establish syllabuses and include methods intended to reduce or eliminate incident risks.
- During task execution, coaches and reviewers observe the task group's work for approximately 30 minutes.

- Coaches evaluate each task group and provide feedback to each member on strengths and areas for improvement.
- These steps are repeated for other groups throughout the drilling campaign.

A summary table of SSC planning and execution is provided below.

Table 2—SSC Planning and Execution Summary

Item	Safe System Circle	Sample of Implementation
Definition	Safety improvement activity conducted by individual or small team (max 5 person)	Sack room operator (including helper and Forklift driver)
Theme	Originated from each individual or team based on activities in their respective area	Avoiding chemical splash while handling liquid chemical
Solving Method	Steps: analyse, observe, hazards identification and possible mitigations	Review JSA, check on suitable PPE availability, observe hazards and available mitigations
Leader	Personnel contributing the theme	Forklift driver
Facilitator	HSSE Coach	HSE Coach using his experience at Shore base and other rigs instruct the team to: 1. Read the JSA and ask everyone about their knowledge on the JSA. 2. Ask to check if the available PPE is suitable or not for the tasks. 3. Inform and explain how incident can be happened. 4. Ask the team to execute the works. 5. Conduct debrief and conclude on suggestions for improvement
Reviewer	Rig HSSE/RSTC and DSM/OIM	Rig HSSE/RSTC and DSM/OIM - to receive HSE Coach explanation about the coaching and debrief results. And after that do require follow up based on debrief meeting
Time	Maximum 30 mins including review document, implement tasks and debrief	HSE Coach to prepare coaching plan based on certain tasks, conducting coaching as needed and prepare report to Rig HSSE/RSTC and DSM/OIM
Evaluation	Individual or small team (max 5 person) able to identify and implement correct mitigations	During debrief HSE Coach need to evaluate team's understanding on the subject being discussed and agree on when he need to comeback to do follow up

Result

Safety culture metric data were systematically collected by HSE coaches and the Offshore HSE team throughout the drilling campaign, utilizing both daily and weekly reports as primary sources. The collected data were subsequently analyzed by the onshore team to refine the implementation of safety measures and ensure alignment with ongoing drilling activities. The data were subjected to statistical analysis and categorized for in-depth evaluation.

Based on the implementation of the Safety Standdown Campaign (SSC), the onshore team concluded the following:

- **Safety Conversations / Leadership Engagement** demonstrated an improvement of over 20% from the first well to the second well. Analysis of the SSC program in the first well identified this metric as a critical mechanism for enhancing HSE communication and providing a platform for the DSM to promote active personnel engagement in HSE initiatives.
- **Hazard Identification, Behavior-Based Safety (BBS) Observations and Reporting** by rig personnel increased by approximately 49%. This improvement was attributed to heightened awareness among the workforce regarding "line of fire" hazards and the potential for work-related injuries, thereby enhancing proactive safety behavior across the team.
- Emergency Drills and Preparedness activities were fully executed, with the rig HSE team reporting 100% completion of all planned drill exercises throughout the drilling campaign. Action plans identified during these exercises were systematically recorded and monitored using an action

tracking tool. Additionally, upon completion of the campaign, an After Action Review (AAR) was conducted by the onshore team to evaluate performance and identify opportunities for improvement.

- **Permit to Work & Risk Assessment Audit** was demonstrated through sample evidence submitted by the Drilling Superintendent and Management (DSM) to the Drilling Superintendent and Drilling Manager. This evidence included PTW audits conducted by both the DSM and Offshore Installation Manager (OIM), with particular focus on high-risk activities. The audits documented findings and corresponding follow-up actions, achieving a compliance rate exceeding 80% relative to the established PTW audit targets
- **Injury/Incident Rates and Near Miss Reporting**, in terms of injury and incident rates, as well as near miss reporting, the drilling campaign was completed without any Lost Time Injuries (LTI). Two dropped object incidents and one near miss were reported, all of which remained within the established HSE Key Performance Indicators (KPIs) for the campaign. These outcomes also effectively addressed management's concerns regarding the implementation of HSE practices on the rig
- **Environmental Program Matrix Implementation** achieved full compliance (100%) by all relevant task groups, including the drilling mud team, galley, site clinic, and rig crew. This success was attributed in part to the support provided by the HSE coach, who delivered targeted guidance and reinforced awareness of applicable government regulations pertaining to offshore drilling operations.
- **Feedback Mechanism Utilization** by the Drilling Superintendent and Management (DSM), through the weekly submission of the Safety Leadership Card with 100% compliance, served as a structured approach to fulfilling the company's Safe System of Work (SSOW) requirements.

Given the diverse backgrounds in nationality, education, and culture among the workforce, it was essential to communicate safety messages and results in a clear and accessible format. The SSC model proved effective in achieving this goal, particularly in the context of a short-term project such as an exploration drilling campaign. The offshore HSE team, supported by HSE coaches, successfully implemented the SSC syllabus. As a result, the initiative contributed significantly to meeting corporate scorecard performance targets while enabling the safe delivery of deepwater exploration wells.

Discussion

As outlined above, various HSE culture tools and analytical approaches have been implemented to enhance the health, safety, and environmental culture within the offshore upstream oil and gas sector. These initiatives aim to minimize operational risks and prevent incidents or accidents. The key challenges and future prospects associated with the application of such innovative approaches in deepwater drilling campaigns are summarized below:

- The process to become a learning organization was faced by the company as the rig entered the company's concession area with a history of incidents, including one potential Lost Time Injury (LTI) and one actual LTI, which raised significant concern. However, the implementation of the Safety Standdown Campaign (SSC) effectively addressed these concerns, demonstrating the program's responsiveness and capability in reinforcing safety culture and risk awareness.
- With the availability of advanced HSE tools designed to enhance safety culture, organizations can effectively adopt and integrate various combinations of these tools to support high-risk operations. For the company, the integration of multiple tools within the framework of the Safety Standdown Campaign (SSC) proved instrumental in achieving the HSE Key Performance Indicators (KPIs) for the drilling campaign and is considered a viable approach for future implementation.

Conclusion

1. In this paper, the applications of SSC for deepwater drilling campaign were reviewed.
2. The application as novel approach in combining QCM, the Company's SSOW and Rig HSE coach program to transform HSE culture in relatively short period of deepwater drilling was also discussed. This approach has been chosen, as it was mandated by the company's management team to implement HSE campaign that will meet the HSE KPI for the company's first drilling campaign.
3. Finally, the challenges and prospects of implementing similar program in future deepwater drilling campaign were discussed. This can provide avenue for future research in this domain for other lifecycles of oil & gas sector such as development and production operations.

Abbreviations

AAR	: After Action Review
BBSO	: Behavior-Based Safety Observation
DMT	: Drilling Management Team
DSM	: Drilling Site Manager
HSE	: Health, Safety, & Environment
HAZID	: Hazard Identification
HOC	: Hazard Observation Card
IBU	: Indonesia Business Unit
JSA	: Job Safety Analysis
KPI	: Key Performance Indicator
LAH	: Look Ahead
LTI	: Lost Time Incidents
OIM	: Offshore Installation Manager
OMS	: Operations Management System
ORF	: Onshore Receiving Facilities
QCM	: Quality Circle Model
PDCA	: Plan Do Check Act
PPE	: Personal Protective Equipment
PTW	: Permit To Work
SSC	: Safe System Circle
SSOW	: Safe System of Work
SWA	: Stop Work Authority

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